Regression and statistical series

The labs and practice sets of Day 4. You type your work in the do-file named on each page; this booklet is the printed copy to read, plan and write notes on. Core first, then Extension, then Challenge.

How the checks work. Every task in the do-file ends with a CHECK line. Run the file with Ctrl + D: a silent CHECK means your answer is correct; red text under a CHECK means look at that task again. Never edit a CHECK line. The solutions booklet explains the reasoning behind each answer; read it after you have tried.

Lab 1: Hours worked

File `C:/NBS_Stata/day4/lab_D4_1.do`
inputs data/lfs_svy.dta
outputs logs/lab_D4_1.log

The situation
The Minister's office asks whether employed women work fewer usual hours than men of the same age, region and education, and whether the public and the private sector differ.

Task 1 Settings, data, flags — Core

The five top lines (diary logs/lab_D4_1.log), open lfs_svy.dta and build the 0/1 flags lf (employed or unemployed) and emp (employed).

Task 2 The design-based gap in usual hours — Core

Estimate mean usual hours of the employed by sex with svy, then the gap men minus women with lincom.

Task 3 The plain model, with a sensible base — Core

Regress usual hours on sex, age group, region and education for the employed. Use Intermediate (code 3) as the education base.

Task 4 Check the model — Core

Check the VIFs, then test for unequal spread.

Task 5 The design-based model — Core

The same model with svy and the employed as the subpopulation.

Task 6 Public and private — Extension

Add the institutional sector (ilo_job1_ins_sector) to the design-based model. Do private-sector employees work more hours?

Task 7 Is the gap the same in both sectors? — Challenge

Let the effect of sex differ by sector (##). Is the interaction significant?

A correct lab. the whole file runs with no red text and ends with "end of do-file".

Lab 2: The analytical note

File `C:/NBS_Stata/day4/lab_D4_2.do`
inputs data/lfs_svy.dta, data/cpi_divisions_long.dta
outputs logs/lab_D4_2.log, output/D4_margins.png, output/D4_infl.png,

The situation
The Minister's office wants two pages by tomorrow: who is more likely to be unemployed, and where inflation stands. Every number in the note must come from this do-file.

Task 1 Settings, data, flags — Core

The five top lines (diary logs/lab_D4_2.log), open lfs_svy.dta and build the 0/1 flags lf and unemp.

Task 2 The design-based model — Core

A design-based logit of unemp on sex, age group and region, for the labour force. Store it as m_svy.

Task 3 Predicted rates and the chart — Core

Predicted unemployment rate by age group; copy r(table) into M at once. Chart it with marginsplot (title, source note) and export it to output/D4_margins.png.

Task 4 Inflation — Core

Open cpi_divisions_long.dta, keep all items (division 0), declare the monthly series, compute year-on-year inflation yoy, chart it and export the chart to output/D4_infl.png. M stays in memory.

Task 5 The note — Core

putdocx: a title, one sentence with the youth rate (100 * M[1, 1], one decimal), the margins chart, one sentence with the latest inflation (yoy[_N], two decimals), the inflation chart. Save it as output/note_q1.docx.

Task 6 The model table — Extension

Reopen lfs_svy.dta first: etable takes the labels from the data in memory. Then export the stored model m_svy with etable to output/model_q1.docx.

Task 7 Which prices drive inflation? — Challenge

Reopen cpi_divisions_long.dta as a panel, compute yoy inside each division, keep August 2026 and the twelve divisions, and sort so the highest inflation is in row 1.

A correct lab. the whole file runs with no red text, ends with "end of do-file", and output/note_q1.docx opens in Word.

Practice sets

File `C:/NBS_Stata/day4/practice_D4.do`. One block per part of the day; each opens its own data, so the parts can be done in any order. Type your number after `nbscheck` where the file asks for it.

Part 1: Tests (lfs_svy.dta, four flags)

P1.1 Core ci means of age for the unemployed. Type the upper limit, two decimals.

P1.2 Core ttest of usual hours, public against private sector (ilo_job1_ins_sector), employed. Type the gap public minus private, two decimals.

P1.3 Core Design-based participation by sex (subpop wap), then the gap men minus women with lincom. Type it, four decimals.

P1.4 Extension Design-based unemployment rate, Praslin & La Digue (6) minus North (1). Type the gap, four decimals.

P1.5 Challenge One test line: are the four age-group unemployment rates all equal? Type the p-value, four decimals.

Part 2: Regression (lfs_svy.dta, four flags)

P2.1 Core Regress usual hours on sex only (employed). Type the coefficient for women, three decimals.

P2.2 Core Add age group. Type R-squared, four decimals.

P2.3 Core Sex, age group and region with Central 2 (3) as the base. Type the coefficient for North, three decimals.

P2.4 Extension Sex, age group, region, education (base 3) and the institutional sector. Type the coefficient for private, three decimals.

P2.5 Challenge After sex, age group, region and education (base 3): how many employed persons have a residual above 40?

Part 3: Diagnostics (lfs_svy.dta, four flags)

P3.1 Core Hours on sex and age group (employed), then estat hettest. Type the p-value, four decimals.

P3.2 Core Same model with vce(cluster psu). Type the SE for women, three decimals.

P3.3 Core Same model with svy, subpop(emp). Type the coefficient for women, three decimals.

P3.4 Extension Merge Less than basic (1) into Basic (2) as edu4 with recode, refit sex, age group, region and ib3.edu4, and type the largest leverage, three decimals.

P3.5 Challenge How many employed persons report more than 80 usual hours?

Part 4: Logit (lfs_svy.dta, four flags)

P4.1 Core Logit of unemp on region only (labour force). Type the coefficient for Praslin & La Digue (6), three decimals.

P4.2 Core margins i.region: type the predicted rate for Perseverance (7), four decimals.

P4.3 Core Design-based logit on sex, age group and region; margins by sex. Type the predicted rate for women, four decimals.

P4.4 Extension Same model with the option or. Type the odds ratio for 55+, three decimals.

P4.5 Challenge Design-based logit on sex, age (c.ilo_age) and region. Predicted rate at age 20, four decimals.

Part 5: Presenting (lfs_svy.dta, four flags; close Word and Excel)

P5.1 Core Store the hours model with sex only as h1 and with age group added as h2 (employed).

P5.2 Core etable of h1 and h2. Type the coefficient for women in h2, three decimals.

P5.3 Core Export the table to output/p5_hours.xlsx.

P5.4 Extension A Word note output/p5_note.docx with a title and one sentence.

P5.5 Challenge Add the design-based marginsplot by age group of Part 4 to a note output/p5_note2.docx.

Part 6: The CPI series (all items, tsset, yoy and mom built)

P6.1 Core Year-on-year inflation for December 2025, two decimals.

P6.2 Core Month-on-month inflation for July 2026, two decimals.

P6.3 Core Average year-on-year inflation over the 12 months of 2025, two decimals.

P6.4 Extension How many months had negative year-on-year inflation?

P6.5 Challenge Inflation over 24 months to August 2026 (L24.), two decimals.

Part 7: The CPI panel (xtset, yoy built)

P7.1 Core Year-on-year inflation of Education (10) in August 2026, two decimals.

P7.2 Core Which division had the lowest inflation in August 2026? Type its code.

P7.3 Core Health (6): the highest year-on-year inflation since 2008, two decimals.

P7.4 Extension How many of the twelve divisions had negative inflation in August 2026?

P7.5 Challenge Chart transport (7) and all items (0) together since 2020 with xtline, overlay.